

Hints & Solutions

#MathBoleTohMathonGo

Q1 (3) - Single Correct

$$\text{Given } \left| \frac{f'(x)}{f''(x)} \cdot \frac{f(x)}{f'(x)} \right| = 0$$

$$\Rightarrow f'(x) \cdot f'(x) - f(x)f''(x) = 0$$

$$\Rightarrow \frac{[f'(x)]^2 - f(x)f''(x)}{[f'(x)]^2} = 0 \Rightarrow \frac{d}{dx} \left[\frac{f(x)}{f'(x)} \right] = 0$$

$$\text{On integrating } \frac{f(x)}{f'(x)} = C \dots (i)$$

$$\text{On putting } x = 0, \frac{f(0)}{f'(0)} = C \Rightarrow C = \frac{1}{2}$$

$$\text{Hence, } \frac{f(x)}{f'(x)} = \frac{1}{2} \dots (ii)$$

$$\text{From Eq. (i), } 2f(x) = f'(x)$$

$$\frac{f'(x)}{f(x)} = 2$$

Again integrating, we have

$$\log[f(x)] = 2x + K$$

$$\text{On putting } x = 0, K = 0$$

$$\therefore f(x) = e^{2x}$$

Q2 (2) - Single Correct

Given,

$$\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \log[f(x)] + g(x) + C$$

$$\text{Now, let } I = \int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx$$

$$= \int \frac{(e^x + \cos x + 1) - (e^x + \sin x + x)}{e^x + \sin x + x} dx$$

$$= \log(e^x + \sin x + x) - x + C$$

$$\therefore f(x) = e^x + \sin x + x \text{ and } g(x) = -x$$

$$\text{Now, } f(x) + g(x) = e^x + \sin x$$

Q3 (3) - Single Correct

$$\text{Let } I = \int x \cdot \frac{\log(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}} dx$$

$$\text{Put } t = \sqrt{x^2 + 1}$$

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned} \Rightarrow dt &= \frac{x}{\sqrt{x^2+1}} dx \\ \therefore I &= \int \log(t + \sqrt{t^2-1}) dx \\ &= \log(t + \sqrt{t^2-1}) \cdot t - \int \frac{1 + \frac{t}{\sqrt{t^2-1}}}{t + \sqrt{t^2-1}} \cdot t dt \\ &= t \cdot \log(t + \sqrt{t^2-1}) - \frac{1}{2} \int \frac{2t}{\sqrt{t^2-1}} dt \\ &= t \cdot \log(t + \sqrt{t^2-1}) - \sqrt{t^2-1} + C \\ &= \sqrt{1+x^2} \log(x + \sqrt{1+x^2}) - x + C \end{aligned}$$

Q4 (1) - Single Correct

$$\begin{aligned} \text{Let } I &= \int \frac{(2x+1)}{(x^2+4x+1)^{3/2}} dx \\ &= \int \frac{(2x+1)}{x^3(1+\frac{4}{x}+\frac{1}{x^2})^{3/2}} dx = \int \frac{2x^{-2}+x^{-3}}{(1+\frac{4}{x}+\frac{1}{x^2})^{3/2}} dx \\ \text{Put } \frac{1}{x^2} + \frac{4}{x} + 1 &= t^2 \\ \Rightarrow \left(\frac{-2}{x^3} + \frac{4}{x^2} \right) dx &= 2t dt \Rightarrow \left(\frac{2}{x^2} + \frac{1}{x^3} \right) dx = -t dt \\ \therefore &= \int \frac{-t}{t^3} dt = \frac{1}{t} + C = \frac{x}{\sqrt{x^2+4x+1}} + C \end{aligned}$$

Q5 (1) - Single Correct

$$\begin{aligned} \text{Let } I &= \int \sin(101x) \sin^{99} x dx \\ &= \int \sin(100x + x) (\sin x)^{99} dx \\ &= \int \{ \sin(100x) \cos x + \cos 100x \sin x \} (\sin x)^{99} dx \\ &= \int \sin(100x) \cos x \cdot (\sin x)^{99} dx + \int \cos(100x) (\sin x)^{100} dx \\ &= \frac{\sin(100x)(\sin x)^{100}}{100} - \frac{100}{100} \int \cos(100x) (\sin x)^{100} dx + \int \cos(100x) (\sin x)^{100} dx \\ &= \frac{\sin(100x)(\sin x)^{100}}{100} + C \end{aligned}$$

Q6 (4) - Single Correct

$$\begin{aligned} \text{Given, } f'(\log x) &= \begin{cases} 1, & 0 < x \leq 1 \\ x, & x > 1 \end{cases} \\ \text{Put } \log x = t &\Rightarrow x = e^t \\ \therefore f'(t) &= \begin{cases} 1, & 0 < e^t \leq 1 \\ e^t, & e^t > 1 \end{cases} \\ \text{On integrating, we get} & \end{aligned}$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$f(t) = \begin{cases} t + C, & 0 < e^t \leq 1 \\ e^t + C, & e^t > 1 \end{cases}$$

$$f(t) = t + C, 0 < e^t \leq 1$$

$$f(0) = 0 + C \Rightarrow C = 0 \quad [\because f(0) = 0]$$

$$\therefore f(t) = t, t \leq 0$$

$$\text{and } f(t) = e^t + C, e^t > 1$$

$$f(t) = e^t = C, t > 0$$

$$f(0) = e^0 + C$$

$$0 = 1 + C \quad [\because f(0) = 0]$$

$$\Rightarrow C = -1$$

$$f(t) = e^t - 1$$

$$\therefore f(t) = \begin{cases} t, & t \leq 0 \\ e^t - 1, & t > 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x, & x \leq 0 \\ e^x - 1, & x > 0 \end{cases}$$

Q7 (3) - Single Correct

$$\text{Let } I = \int \frac{px^{p+2q-1} - qx^{q-1}}{(x^{p+q} + 1)^2} dx$$

$$= \int \frac{px^{p-1} - qx^{-q-1}}{(x^p + x^{-q})^2} dx$$

On taking x^q as x^{2q} common from denominator and take it in numerator.

$$\text{Put } x^p + x^{-q} = t$$

$$\Rightarrow (px^{p-1} - qx^{-q-1}) dx = dt$$

$$\therefore I = \int \frac{dt}{t^2} = -\frac{1}{t} + C = -\left(\frac{x^q}{x^{p+q}+1}\right) + C$$

Q8 (3) - Single Correct

$$\text{Let } I_n = \int \sin^n x dx$$

$$= \int \sin^{n-1} x \sin x dx$$

$$= -\sin^{n-1} x \cos x + \int (n-1) \sin^{n-2} x \cos^2 x dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int (\sin^{n-2} x - \sin^n x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1)I_{n-2} - (n-1)I_n$$

$$\therefore nI_n = -\sin^{n-1} x \cos x + (n-1)I_{n-2} + C \dots(i)$$

Put $n = 6$ in Eq. (i), we get

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$6I_6 = -\sin^5 x \cos x + 5I_4 + C$$

$$\Rightarrow 5I_4 - 6I_6 = \sin^5 x \times \cos x + C \dots (ii)$$

$$\text{Now, } \sin^5 x \cos x = \frac{1}{2} \sin^4 x (2 \sin x \cos x)$$

$$= \frac{1}{2} \sin^4 x \sin 2x = \frac{1}{2} \left[\frac{1 - \cos 2x}{2} \right]^2 \sin 2x$$

$$\Rightarrow \sin^5 x \cdot \cos x = \frac{\sin 2x}{8} [1 + \cos^2 2x - 2 \cos 2x] \dots (iii)$$

From Eqs. (ii) and (iii), we get

$$5I_4 - 6I_6 = \frac{\sin 2x}{8} [1 + \cos^2 2x - 2 \cos 2x] + C$$

Q9 (3) - Single Correct

We know that

$$\sec^{-1} \sqrt{1+x^2} = \tan^{-1} x$$

$$\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2 \tan^{-1} x, \text{ for } x > 0$$

$$I = \int \frac{e^{\tan^{-1} x}}{1+x^2} \left\{ (\tan^{-1} x)^2 + 2 \tan^{-1} x \right\} dx$$

$$\text{Put } \tan^{-1} x = t \Rightarrow \frac{1}{1+x^2} dx = dt$$

$$\therefore I = \int e^t (t^2 + 2t) dt = e^t \cdot t^2 + C$$

$$= e^{\tan^{-1} x} (\tan^{-1} x)^2 + C$$

$$= e^{\tan^{-1} x} \cdot \left[\left(\sec^{-1} \sqrt{1+x^2} \right) \right]^2 + C$$

Q10 (1) - Single Correct

$$\text{Let } I = \int \prod_{r=1}^n (x+r) \left(\sum_{k=1}^n \frac{1}{x+k} \right) dx$$

$$\text{Now, consider } A = (x+1)(x+2)(x+3) \cdots (x+n)$$

$$A = \prod_{r=1}^n (x+r) \dots (i)$$

$$\text{On taking log both sides, we have } \log A = \log(x+1) + \log(x+2) + \cdots + \log(x+n)$$

On differentiating w.r.t. x , we get

$$\frac{1}{A} \frac{dA}{dx} = \frac{1}{x+1} + \frac{1}{x+2} + \frac{1}{x+3} + \cdots + \frac{1}{x+n}$$

$$\Rightarrow \frac{dA}{dx} = A \left[\frac{1}{x+1} + \frac{1}{x+2} + \frac{1}{x+3} + \cdots + \frac{1}{x+n} \right] = A \left[\sum_{k=1}^n \frac{1}{x+k} \right]$$

$$\Rightarrow \frac{dA}{dx} = \prod_{r=1}^n (x+r) \left[\sum_{k=1}^n \frac{1}{x+k} \right]$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

[using Eq.

On integrating both sides, we have,

$$A = \int \prod_{r=1}^n (x+r) \left(\sum_{k=1}^n \frac{1}{x+k} \right) dx$$

$$\Rightarrow \int \prod_{r=1}^n (x+r) \left(\sum_{k=1}^n \frac{1}{x+k} \right) dx = \prod_{r=1}^n (x+r)$$

Q11 (2) - Single Correct

$$\text{Let } I_n = \int \cot^n x dx$$

$$= \int \cot^{n-2} x (\operatorname{cosec}^2 x - 1) dx = -\frac{u^{n-1}}{n-1} - I_{n-2}$$

$$\Rightarrow I_n + I_{n-2} = -\frac{u^{n-1}}{n-1} \quad [\text{put } n = 2, 3, 4, \dots, 10]$$

$$I_2 + I_0 = -\frac{u}{1}$$

$$I_3 + I_1 = -\frac{u^2}{2}$$

$$I_4 + I_2 = -\frac{u^3}{3}$$

$$\vdots$$

$$I_{10} + I_8 = -\frac{u^9}{9}$$

On adding all the above, we get

$$10 + I_1 + 2(I_2 + I_3 + \dots + I_8) + I_5 + I_{10} = -\left(u + \frac{u^2}{2} + \dots + \frac{u^9}{9}\right)$$

Q12 (3) - Single Correct

$$\text{Let } I = \int \frac{x^2(1-\log x)}{\log^4 x - x^4} dx$$

$$= \int \frac{x^2(1-\log x)}{x^4 \left[\left(\frac{\log x}{x} \right)^4 - 1 \right]} dx = \int \frac{1-\log x}{x^2 \left[\left(\frac{\log x}{x} \right)^4 - 1 \right]} dx$$

$$\text{Put } \frac{\log x}{x} = t$$

$$\Rightarrow \left(\frac{1-\log x}{x^2} \right) dx = dt$$

$$\therefore I = \int \frac{dt}{(t^4-1)} = \int \frac{dt}{(t^2+1)(t^2-1)}$$

$$= \frac{1}{2} \int \frac{(t^2+1) - (t^2-1)}{(t^2+1)(t^2-1)} dt$$

$$= \frac{1}{2} \left[\int \frac{dt}{t^2-1} - \int \frac{dt}{t^2+1} \right]$$

$$= \frac{1}{2} \left(\frac{1}{2} \log \frac{t-1}{t+1} - \tan^{-1} t \right) + C$$

$$= \frac{1}{4} \log \left(\frac{\log x - x}{\log x + x} \right) - \frac{1}{2} \tan^{-1} \left(\frac{\log x}{x} \right) + C$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Q13 (4) - Single Correct

On solving

$$\frac{\sin \theta}{\cos 3\theta} = \frac{1}{2} [\tan 3\theta - \tan \theta]$$

$$\frac{\sin 3\theta}{\cos 9\theta} = \frac{1}{2} [\tan 9\theta - \tan 3\theta]$$

$$\frac{\sin 9\theta}{\cos 27\theta} = \frac{1}{2} [\tan 27\theta - \tan 9\theta]$$

$$\therefore \int \left(\frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta} \right) d\theta$$

$$= \frac{1}{2} \int (\tan 27\theta - \tan \theta) d\theta$$

$$= \frac{1}{2} \left[\frac{1}{27} \log(\sec 27\theta) - \log \sec \theta \right] + C$$

Q14 (1) - Single Correct

$$\text{Let } I = \int \frac{x^2+20}{(x \sin x + 5 \cos x)^2} dx$$

$$= \int \frac{x^8(x^2+20)}{x^8(x \sin x + 5 \cos x)^2} dx$$

$$= \int \frac{x^8(x^2+20)}{(x^5 \sin x + 5x^4 \cos x)^2} dx$$

$$= \int \frac{(x^5+20x^3) \cos x}{(x^5 \sin x + 5x^4 \cos x)^2} \cdot \frac{x^5}{\cos x} dx$$

$$\text{Note that } \frac{d}{dx} [x^5 \sin x + 5x^4 \cos x] = (x^5 + 20x^3) \cos x$$

Integrating by parts, we have

$$I = -\frac{x^5}{\cos x (x^5 \sin x + 5x^4 \cos x)} + \int \frac{5x^4 \cos x + x^5 \sin x}{x^5 \sin x + 5x^4 \cos x} \sec^2 x dx$$

$$= -\frac{x}{\cos x (x \sin x + 5 \cos x)} + \tan x + C$$

Q15 (3) - Single Correct

$$\text{Let } I = n! \int dx - n! \int \frac{\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!}\right)}{1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}} dx$$

$$= n!(x) - n! \log \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) + C$$

$$= n! \left[\log_e e^x - \log_e \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \right] + C$$

$$= n! \log_e \left(\frac{e^x}{1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}} \right) + C$$

Q16 (3, 4) - Multiple Correct

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Given $\int e^{3x} \cos 4x dx = e^{3x}(A \sin 4x + B \cos 4x) + C$

Now, $\int e^{3x} \cos 4x = \frac{e^{3x}}{(3)^2 + (4)^2} [4 \sin 4x + 3 \cos 4x] + C$

$$= \frac{e^{3x}}{25} [4 \sin 4x + 3 \cos 4x] + C$$

$$= e^{3x} \left[\frac{4}{25} \sin 4x + \frac{3}{25} \cos 4x \right] + C$$

$$\Rightarrow A = \frac{4}{25}, B = \frac{3}{25}$$

$$\therefore 3A = 3 \times \frac{4}{25} = \frac{3}{25} \times 4 = 4B$$

$$\text{and } 4A + 3B = 4 \times \frac{4}{25} + 3 \times \frac{3}{25}$$

$$= \frac{16}{25} + \frac{9}{25} = \frac{25}{25} = 1$$

Q17 (2, 3, 4) - Multiple Correct

$$f'(x) = 3x^2 \cdot \sin(1/x) - x \cos(1/x), x \neq 0$$

On integrating, $f(x) = \begin{cases} x^3 \cdot \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$

$$\therefore f(x) \text{ is continuous at } x = 0 \Rightarrow f(x) \text{ is differentiable at } x = 0 \text{ Again, } f'(x) = 3x^2 \cdot \sin(1/x) - x \cos(1/x)$$

$$\Rightarrow \text{Continuous at } x = 0$$

$$\therefore f'(x) \text{ is not differentiable at } x = 0$$

Q18 (1, 2, 3) - Multiple Correct

We have, $\int x e^x \cdot \sin x dx = \text{Imaginary part of } \int x e^{(1+i)x} dx$

$$= \text{Im} \int x e^{(1+i)x} dx, \text{ using integration by parts}$$

$$= \text{Im} \left\{ \frac{x e^{(1+i)x}}{(1+i)} - \frac{e^{(1+i)x}}{(1+i)^2} \right\}$$

$$= \frac{x e^x}{2} \cdot \text{Im}(\cos x + i \sin x)(1-i) - \frac{e^x}{4} \text{Im}[\cos x + i \sin x \cdot x(-2i)]$$

$$= \frac{e^x}{2} [x \cdot (\sin x - \cos x) + \cos x] + C$$

Q19 (1, 3) - Multiple Correct

Given, $u = \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [b \sin bx + a \cos bx]$

and $v = \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$

Now, $u^2 = \frac{e^{2ax}}{(a^2 + b^2)^2} (b \sin bx + a \cos bx)^2$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\text{and } v^2 = \frac{e^{2ax}}{(a^2+b^2)^2} (a \sin bx - b \cos bx)^2$$

$$\text{Then, } u^2 + v^2 = \frac{e^{2ax}}{(a^2+b^2)^2} (a^2 + b^2)$$

$$\Rightarrow u^2 + v^2 = \frac{e^{2ax}}{a^2+b^2}$$

$$\Rightarrow (a^2 + b^2) (u^2 + v^2) = e^{2ax}$$

$$\text{Again, } u^2 - v^2$$

$$= \frac{e^{2ax} [(b \sin bx + a \cos bx + a \sin bx - b \cos bx) (b \sin bx + a \cos bx - a \sin bx + b \cos bx)]}{(a^2+b^2)^2}$$

$$= \frac{e^{2ax}}{(a^2+b^2)^2} [\{(a+b) \sin bx + (a-b) \cos bx\} (b-a) \sin bx + (a+b) \cos bx]$$

$$= \frac{e^{2ax} [- (a^2-b^2) \sin^2 bx + (a+b)^2 \sin bx \cos bx - (a-b)^2 \sin bx \cos bx + (a^2-b^2) \cos^2 bx]}{(a^2+b^2)^2}$$

$$\Rightarrow u^2 - v^2 \neq e^{2ax}$$

$$\text{Now, } \frac{v}{u} = \frac{a \sin bx - b \cos bx}{a \cos bx + b \sin bx}$$

$$\Rightarrow \frac{v}{u} = \frac{\tan bx - b/a}{1 + (b/a) \tan bx}$$

$$\Rightarrow \frac{v}{u} = \tan \left[bx - \tan^{-1} \left(\frac{b}{a} \right) \right]$$

$$\therefore \tan^{-1} \left(\frac{v}{u} \right) = bx - \tan^{-1} \left(\frac{b}{a} \right)$$

$$\text{i. } \theta. \quad \tan^{-1} \left(\frac{v}{u} \right) + \tan^{-1} \left(\frac{b}{a} \right) = bx$$

Q20 (1, 2, 4) - Multiple Correct

$$\text{Let } I = \int \sqrt{1 + 2 \tan x (\sec x + \tan x)} dx$$

$$= \int \sqrt{1 + 2 \tan x \sec x + 2 \tan^2 x} dx$$

$$= \int \sqrt{[(1 + \tan^2 x) + \tan^2 x + 2 \tan x \sec x]} dx$$

$$= \int \sqrt{[\sec^2 x + \tan^2 x + 2 \tan x \sec x]} dx$$

$$= \int \sqrt{(\sec x + \tan x)^2} dx = \int (\sec x + \tan x) dx$$

$$= \log |\sec x + \tan x| + \log |\sec x| + C$$

So, option (2) is correct.

$$= \log \frac{1}{\sec x - \tan x} + \log |\sec x| + C$$

$$= -\log |\sec x - \tan x| + \log |\sec x| + C$$

So, option (1) is correct.

Hints & Solutions

#MathBoleTohMathonGo

$$\text{Again, } \log |\sec x + \tan x| + \log |\sec x| + C$$

$$= \log |\sec x (\sec x + \tan x)| + C$$

$$= \log |\sec^2 x + \sec x \tan x| + C$$

$$= \log |1 + \tan^2 x + \sec x \tan x| + C$$

$$= \log |1 + \tan x (\tan x + \sec x)| + C$$

So, option (4) is correct.

Q21 (2, 4) - Multiple Correct

$$\text{Let } I = \int \frac{1}{x^2-1} \log \left| \frac{x-1}{x+1} \right| dx$$

$$\text{Put } \log \left| \frac{x-1}{x+1} \right| = t$$

$$\Rightarrow \frac{x+1}{x-1} \cdot \frac{2}{(x+1)^2} dx = dt$$

$$\Rightarrow \left(\frac{2}{x^2-1} \right) dx = dt$$

$$\therefore I = \frac{1}{2} \int t \cdot dt = \frac{1}{2} \frac{t^2}{2} + C$$

$$= \frac{1}{4} \log^2 \left| \frac{x-1}{x+1} \right| + C$$

So, option (2) is correct.

$$= \frac{1}{4} \left\{ -\log \left| \frac{x+1}{x-1} \right| \right\}^2 + C$$

$$= \frac{1}{4} \log^2 \left| \frac{x+1}{x-1} \right| + C$$

So, option (4) is correct.

Q22 (1, 2, 3, 4) - Multiple Correct

$$\text{Let } I = \int \frac{dx}{\sqrt{x-x^2}} = \int \frac{dx}{\sqrt{-(x^2-x+\frac{1}{4}-\frac{1}{4})}}$$

$$= \int \frac{dx}{\sqrt{-(x-\frac{1}{2})^2 - (\frac{1}{2})^2}}$$

$$= \int \frac{dx}{\sqrt{(\frac{1}{2})^2 - (x-\frac{1}{2})^2}}$$

$$= \sin^{-1} \frac{(x-\frac{1}{2})}{1/2} + C$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$= \sin^{-1}(2x - 1) + C \text{ [option 2]}$$

$$= \cos^{-1}(2\sqrt{x - x^2}) + C \text{ [option 4]}$$

$$\text{Again, } l = \int \frac{dx}{\sqrt{x-x^2}} = \int \frac{dx}{\sqrt{x}\sqrt{1-x}}$$

$$\Rightarrow 1 = \int \frac{dx}{\sqrt{x}\sqrt{(1)^2 - (\sqrt{x})^2}}$$

$$\text{Put } \sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$\therefore l = 2 \int \frac{dt}{\sqrt{1^2 - t^2}}$$

$$= 2 \sin^{-1} t + C$$

$$= 2 \cdot \sin^{-1} \sqrt{x} + C \text{ [option 1]}$$

$$= \cos^{-1}(1 - 2x) + C$$

$$= -\cos^{-1}(2x - 1) + C \text{ [option 3]}$$

Q23 (1, 2, 3) - Multiple Correct

$$\text{Let } I = \int \frac{\sin 2x}{(\sin^4 x + \cos^4 x)} dx$$

$$= \int \frac{2 \sin x \cos x}{\cos^4 x} dx$$

$$= \int \frac{2 \tan x \cdot \sec^2 x}{(\tan^2 x)^2 + 1} dx$$

$$\text{Put } \tan^2 x = t$$

$$\Rightarrow 2 \tan x \cdot \sec^2 x dx = dt$$

$$\therefore = \int \frac{dt}{t^2 + 1} = \tan^{-1} t + C$$

$$= \tan^{-1}(\tan^2 x) + C \text{ [option c]}$$

$$= \frac{\pi}{2} - \cot^{-1}(\tan^2 x) + C$$

$$[\text{using } \tan^{-1} x + \cot^{-1} x = \pi/2]$$

$$= -\cot^{-1}(\tan^2 x) + C \text{ [option b]}$$

$$\text{Also, } \tan^{-1}(x) = \cot^{-1}(1/x)$$

$$\Rightarrow \tan^{-1}(\tan^2 x) = \cot^{-1}(\cot^2 x) + C \text{ [option 1]}$$

$$\text{and } -\tan^{-1}(\cos 2x) + C = -\tan^{-1}\left(\frac{1 - \tan^2 x}{1 + \tan^2 x}\right) + C$$

$$= -\tan^{-1} 1 + \tan^{-1}(\tan^2 x) + C$$

$$= \tan^{-1}(\tan^2 x) + C$$

Hints & Solutions

#MathBoleTohMathonGo

Q24 (1) - Multiple Correct

$$\begin{aligned}\text{Let } I &= \int \frac{dx}{x^r(1+x^r)^{1/r}} = \int \frac{dx}{x^{r+1}\left(1+\frac{1}{x^r}\right)^{1/r}} \\ &= \int (1+x^{-r})^{-1/r} \cdot x^{-r-1} dx = -\frac{1}{r} \int (1+x^{-r})^{\frac{-1}{r}} d(1+x^{-r}) \\ &= -\frac{1}{r} \frac{(1+x^{-r})^{-\frac{1}{r}+1}}{\left(-\frac{1}{r}+1\right)} + C = \frac{1}{1-r} \left(1+\frac{1}{x^r}\right)^{\frac{r-1}{r}} + C \dots (i)\end{aligned}$$

Now, put $r = 3$ in Eq. (i), we get

$$I = \frac{1}{-2} (1+x^{-3})^{2/3} + C$$

$$\text{But } I = \alpha(1+x^{-3})^{\beta} + C \Rightarrow \alpha = -\frac{1}{2}, \beta = \frac{2}{3}$$

$$\therefore \text{Required equation is } x^2 - \left(-\frac{1}{2} + \frac{2}{3}\right)x + \left(-\frac{1}{2}\right)\left(\frac{2}{3}\right) = 0$$

$$\Rightarrow x^2 - \frac{1}{6}x - \frac{1}{3} = 0$$

$$\Rightarrow 6x^2 - x - 2 = 0$$

Q25 (1) - Multiple Correct

$$\text{Let } I = \int \frac{(x^5+x^4)}{\sqrt{5x^6+6x^5+1}} dx$$

$$\text{Put } 5x^6 + 6x^5 + 1 = t$$

$$\Rightarrow (30x^5 + 30x^4) dx = dt \Rightarrow (x^5 + x^4) dx = \frac{dt}{30}$$

$$\therefore I = \frac{1}{30} \int \frac{dt}{t^{1/2}} = \frac{1}{30} \cdot \frac{t^{1/2}}{1/2} + C$$

$$= \frac{1}{15} \sqrt{5x^6 + 6x^5 + 1} + C$$

Q26 (4) - Multiple Correct

We know that

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \text{ if } -1 \leq x \leq 1 \text{ and } x \geq 3$$

$$\therefore \sin^{-1}(\log x) + \cos^{-1}(\log x) \text{ does not exist.}$$

Hence, the given integral does not exist.

Q27 (1) - Multiple Correct

$$\text{Let } I = \int \frac{\operatorname{cosec}^2 x - 2011}{(\cos x)^{2011}} dx$$

$$\int (\cos x)^{-2011} \operatorname{cosec}^2 x dx - \int \frac{2011}{(\cos x)^{2011}} dx$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned}
 &= (\cos x)^{-2011}(-\cot x) - \int (-2011)(\cos x)^{-2012}(-\sin x)(-\cot x)dx - 2011 \int \frac{dx}{(\cos x)^{2011}} \\
 &= (\cos x)^{-2011}(-\cot x) + 2011 \int \frac{dx}{(\cos x)^{2011}} - 2011 \int \frac{dx}{(\cos x)^{2011}} + C \\
 &= \frac{-\cot x}{(\cos x)^{2011}} + C
 \end{aligned}$$

Q28 (3) - Multiple Correct

Given,

$$P = \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (b \sin bx + a \cos bx) + C_1$$

$$Q = \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C_2$$

Now, $P + iQ$

$$= \frac{e^{ax}}{a^2 + b^2} [(b \sin bx + a \cos bx) + i(a \sin bx - b \cos bx)] + C_1 + C_2$$

$$= \frac{e^{ax}}{a^2 + b^2} [(b \sin bx + a \cos bx) + i(a \sin bx - b \cos bx)] + C \text{ [where, } C = C_1 + C_2]$$

Given that, $P + iQ = Z + C$

$$\Rightarrow Z = \frac{e^{ax}}{a^2 + b^2} [(b \sin bx + a \cos bx) + i(a \sin bx - b \cos bx)]$$

$$\therefore |Z| = \frac{e^{ax}}{a^2 + b^2} \sqrt{(b \sin bx + a \cos bx)^2 + (a \sin bx - b \cos bx)^2}$$

$$= \frac{e^{ax}}{a^2 + b^2} \sqrt{\begin{bmatrix} b^2 \sin^2 bx + a^2 \cos^2 bx \\ + 2ab \sin bx \cos bx + a^2 \sin^2 bx \\ + b^2 \cos^2 bx - 2ab \sin bx \cos bx \end{bmatrix}}$$

$$= \frac{e^{ax}}{a^2 + b^2} \sqrt{a^2 + b^2} = \frac{e^{ax}}{\sqrt{a^2 + b^2}}$$

Q29 (2) - Paragraph 1

$$I_{n,-m} = \int \frac{\sin^n x}{\cos^m x} dx$$

$$= \int \sin^{n-1} x \frac{\sin x}{\cos^m x} dx$$

$$= \sin^{n-1} x \cdot \frac{(\cos x)^{-m+1}}{m-1} - \int (n-1) \sin^{n-2} x \cos x \frac{(\cos x)^{-m+1}}{m-1} dx$$

$$= \frac{1}{m-1} \cdot \frac{\sin^{n-1} x}{\cos^{m-1} x} - \frac{n-1}{m-1} \int \frac{\sin^{n-2} x}{\cos^{m-2} x} dx$$

$$I_{n,-m} = \frac{1}{m-1} \cdot \frac{\sin^{n-1} x}{\cos^{m-1} x} - \frac{n-1}{m-1} I_{n-2,2-m}$$

$$\Rightarrow I_{n,-m} + \frac{n-1}{m-1} I_{n-2,2-m} = \frac{1}{m-1} \cdot \frac{\sin^{n-1} x}{\cos^{m-1} x}$$

Q30 (4) - Paragraph 1

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned}
 \text{Let } u_n &= \int \frac{x^n}{\sqrt{ax^2+2bx+c}} \\
 \therefore u_{n+1} &= \int \frac{x^{n+1}}{\sqrt{ax^2+2bx+c}} = \int \frac{x^n \cdot x}{\sqrt{ax^2+2bx+c}} \\
 \Rightarrow u_{n+1} &= \frac{1}{2a} \int \frac{x^n(2ax+2b-2b)}{ax^2+bx+c} dx \\
 \Rightarrow 2au_{n+1} &= \int \frac{x^n(2ax+2b)}{\sqrt{ax^2+2bx+c}} dx - 2b \int \frac{x^n}{\sqrt{ax^2+2bx+c}} dx \\
 &= x^n \cdot 2\sqrt{ax^2+2bx+c} - 2n \int x^{n-1} \sqrt{ax^2+2bx+c} - 2bu_n \\
 &= 2x^n \sqrt{ax^2+2bx+c} - 2n \int \frac{x^{n-1}(ax^2+2bx+c)}{\sqrt{ax^2+2bx+c}} - 2bu_n \\
 &= 2x^n \sqrt{ax^2+2bx+c} - 2n (au_{n-1} - 2bu_n + cu_{n-1}) - 2bu_n \\
 \Rightarrow 2a(n+1)u_{n+1} &= 2x^n \sqrt{ax^2+2bx+c} - 4nbu_n - 2nCu_{n-1} - 1 - 2bu_n \\
 \Rightarrow x^n \sqrt{ax^2+2bx+c}
 \end{aligned}$$

Q31 (3) - Paragraph 2

Given, $x = e^t \cos t$ and $y = e^t \sin t$

$$\begin{aligned}
 \Rightarrow \frac{dx}{dt} &= e^t \cos t + (-\sin t)e^t \\
 &= e^t(\cos t - \sin t) \\
 \text{and } \frac{dy}{dt} &= e^t \sin t + \cos t e^t \\
 &= e^t(\cos t + \sin t)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{\cos t + \sin t}{\cos t - \sin t} \\
 \Rightarrow \frac{dy}{dx} &= \frac{1+\tan t}{1-\tan t} = \tan\left(\frac{\pi}{4} + t\right)
 \end{aligned}$$

Since, α is the angle between the tangent to the curve and X -axis, then

$$\begin{aligned}
 \frac{dy}{dx} &= \tan \alpha \\
 \Rightarrow \tan \alpha &= \tan\left(\frac{\pi}{4} + t\right) \Rightarrow \alpha = \frac{\pi}{4} + t \\
 \therefore t &= \alpha - \frac{\pi}{4}
 \end{aligned}$$

Q32 (2) - Paragraph 2

$$\begin{aligned}
 \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{\cos t + \sin t}{\cos t - \sin t} \right) \cdot \frac{dt}{dx} \\
 &= \frac{[(\cos t - \sin t)(-\sin t + \cos t) - (\cos t + \sin t)(-\sin t - \cos t)]}{(\cos t - \sin t)^2} \times \frac{1}{e^t(\cos t - \sin t)}
 \end{aligned}$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(\cos t - \sin t)^2 + (\cos t + \sin t)^2}{(\cos t - \sin t)^2} \times \frac{1}{e^t(\cos t - \sin t)}$$

$$= \frac{2}{(\cos t - \sin t)^2} \times \frac{1}{e^t(\cos t - \sin t)}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{2}{e^t} \times \frac{1}{(\cos t - \sin t)^3}$$

$$\therefore \left[\frac{d^2y}{dx^2} \right]_{t=0} = 2$$

Q33 (2) - Paragraph 3

We have,

$$\int (x^3 - 2x^2 + 3x - 1) \cos 2x dx$$

$$= (x^3 - 2x^2 + 3x - 1) \cdot \left(\frac{\sin 2x}{2} \right) - (3x^2 - 4x + 3) \cdot \left(-\frac{\cos 2x}{4} \right)$$

$$+ (6x - 4) \cdot \left(-\frac{\sin 2x}{8} \right) - (6) \cdot \left(\frac{\cos 2x}{16} \right) + C$$

$$= \frac{\sin 2x}{4} (2x^3 - 4x^2 + 6x - 2 - 3x + 2) + \frac{\cos 2x}{8} (6x^2 - 8x + 6 - 3) + C$$

$$= \frac{\sin 2x}{4} (2x^3 - 4x^2 + 3x) + \frac{\cos 2x}{8} (6x^2 - 8x + 3) + C$$

$$\therefore u(x) = 2x^3 - 4x^2 + 3x \text{ and } v(x) = 6x^2 - 8x + 3$$

Q34 (3) - Paragraph 3

We have,

$$\int x^4 \cdot e^{2x} dx = (x^4) \cdot \left(\frac{e^{2x}}{2} \right) - (4x^3) \left(\frac{e^{2x}}{4} \right)$$

$$+ (12x^2) \cdot \left(\frac{e^{2x}}{8} \right) - (24x) \left(\frac{e^{2x}}{16} \right) + (24) \left(\frac{e^{2x}}{32} \right) + C$$

$$= \frac{e^{2x}}{2} \left[x^4 - 2x^3 + 3x^2 - 3x + \frac{3}{2} \right] + C$$

Q35 (2) - Paragraph 4

$$\text{For } a = 0, L = \lim_{x \rightarrow \infty} \frac{\int \sin^2 x dx}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{\int (1 - \cos 2x) dx}{2x} = \lim_{x \rightarrow \infty} \left[\frac{x}{2x} - \frac{\sin 2x}{4x} \right]$$

$$= \lim_{x \rightarrow \infty} \left[\frac{1}{2} - \frac{\sin 2x}{4x} \right] = \frac{1}{2}$$

Q36 (4) - Paragraph 4

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned} \text{For } a = 1, L &= \lim_{x \rightarrow \infty} \frac{\int 4 \sin^2 x dx}{x} \\ &= \lim_{x \rightarrow \infty} \frac{2 \int (1 - \cos 2x) dx}{x} = \frac{2 \left(x - \frac{\sin 2x}{2} \right)}{x} \\ &= \lim_{x \rightarrow \infty} 2 - \frac{\sin 2x}{x} = 2 \end{aligned}$$

Q37 (1) - Integer Type

$$\begin{aligned} \text{Given, } g(x) &= \int x^{27} (1 + x + x^2)^6 (6x^2 + 5x + 4) dx \\ &= \int (x^4 + x^5 + x^6)^6 (6x^5 + 5x^4 + 4x^3) dx \end{aligned}$$

$$\text{Let } x^6 + x^5 + x^4 = t$$

$$\Rightarrow (6x^5 + 5x^4 + 4x^3) dx = dt$$

$$g(x) = \int t^6 dt = \frac{t^7}{7} + C$$

$$= \frac{1}{7} (x^4 + x^5 + x^6)^7 + C$$

$$g(0) = 0 \text{ [since, } g(x) \text{ is pass through the origin]}$$

$$\Rightarrow 0 = 0 + C \Rightarrow C = 0$$

$$\text{Now, } g(x) = \frac{1}{7} (x^4 + x^5 + x^6)^7$$

$$g(-1) = \frac{1}{7} (1 - 1 + 1)^7 = \frac{1}{7}$$

$$\text{Hence, } |7g(-1)| = 7 \times \frac{1}{7} = 1$$

Q38 (4) - Integer Type

$$\text{Given, } f(xy) = f(x) \cdot f(y) \dots (i)$$

$$\text{On putting, } x = y = 1, f(1) = 1$$

On differentiating w.r.t. x and y as constant

$$f'(xy) \cdot y = f'(x) f(y)$$

$$\text{When put } x = 1, \text{ then } f'(y)y = f'(1) \cdot f(y)$$

$$\Rightarrow \frac{f'(y)}{f(y)} = \frac{f'(1)}{y}$$

$$\therefore \frac{f'(x)}{f(x)} = \frac{x}{f'(1)} \dots (ii)$$

$$\text{Here, } f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{1 + h\{1 + g(h)\} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \{1 + g(h)\} = 2$$

From Eq. (ii), we get

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\frac{f(x)}{f'(x)} = \frac{x}{2}$$

$$\therefore \int \frac{f(x)}{f'(x)} dx = \frac{x^2}{4} + C$$

On comparing, $k = 4$

Q39 (1) - Integer Type

Given,

$$\int e^{x \sin x + \cos x} \left(\frac{x^4 \cos^3 x - x \sin x + \cos x}{x^2 \cos^2 x} \right) dx$$

$$= e^{x \sin x + \cos x} \frac{f(x)}{x} + C$$

$$\text{Now, } \int e^{x \sin x + \cos x} \left(\frac{x^4 \cos^3 x - x \sin x + \cos x}{x^2 \cos^2 x} \right) dx$$

$$= \int x \cdot e^{(x \sin x + \cos x)} \cdot x \cos x dx - \int e^{(x \sin x + \cos x)} \cdot \frac{x \sin x - \cos x}{(x \cos x)^2} dx$$

On applying integration by parts, we get

$$= \left\{ x \cdot e^{x \sin x + \cos x} - \int e^{(x \sin x + \cos x)} dx \right\} - \left\{ e^{(x \sin x + \cos x)} \frac{1}{x \cos x} - \int e^{(x \sin x + \cos x)} dx \right\}$$

$$= e^{x \sin x + \cos x} \left(x - \frac{1}{x \cos x} \right) + C$$

$$\Rightarrow f(x) = \frac{x^2 \cos x - 1}{\cos x}$$

$$\therefore f(0) = 1 \Rightarrow |f(0)| = 1$$

Q40 (4) - Integer Type

$$\text{Given, } A = \int \frac{dx}{x^2 + 6x + 25} \text{ and } B = \int \frac{dx}{x^2 - 6x - 27}$$

$$\Rightarrow A = \int \frac{dx}{x^2 + 6x + 9 - 9 + 25}$$

$$\text{and } B = \int \frac{dx}{x^2 - 6x + 9 - 9 - 27}$$

$$\Rightarrow A = \int \frac{dx}{(x+3)^2 + 16} \text{ and } B = \int \frac{dx}{(x-3)^2 - 36}$$

$$\Rightarrow A = \int \frac{dx}{(x+3)^2 + (4)^2} \text{ and } B = \int \frac{dx}{(x-3)^2 - (6)^2}$$

$$\Rightarrow A = \frac{1}{4} \tan^{-1} \left(\frac{x+3}{4} \right) \text{ and } B = \frac{1}{2 \times 6} \log \left| \frac{x-3-6}{x-3+6} \right|$$

$$\text{Now, } 12(A + B) = 12 \left[\frac{1}{4} \tan^{-1} \left(\frac{x+3}{4} \right) + \frac{1}{12} \log \frac{x-9}{x+3} \right]$$

$$= 3 \tan^{-1} \left(\frac{x+3}{4} \right) + \log \left| \frac{x-9}{x+3} \right|$$

$$\therefore \lambda = 3, \mu = 1 \Rightarrow \lambda + \mu = 4$$

Q41 (2) - Integer Type

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\text{Given, } \int \frac{dx}{x^{2003}+x} = \frac{1}{p} \log\left(\frac{x^q}{1+x^r}\right) + C$$

$$\text{Let } I = \int \frac{dx}{x^{2008}+x} = \int \frac{dx}{x^{2008}\left(1+\frac{1}{x^{2007}}\right)}$$

$$\text{Put } 1 + \frac{1}{x^{2007}} = t \Rightarrow -\frac{2007}{x^{2008}} dx = dt$$

$$\therefore I = -\frac{1}{2007} \int \frac{dt}{t}$$

$$I = -\frac{1}{2007} \int \frac{dt}{t}$$

$$= -\frac{1}{2007} \log|t| + C$$

$$= -\frac{1}{2007} \log\left|1 + \frac{1}{x^{2007}}\right| + C$$

$$= -\frac{1}{2007} \log\left|\frac{x^{2007} + 1}{x^{2007}}\right| + C$$

$$= \frac{1}{2007} \log\left|\frac{x^{2007}}{1 + x^{2007}}\right| + C$$

$$\Rightarrow p = 2007, q = 2007 \text{ and } r = 2007$$

$$\text{Hence, } \frac{p+q}{r} = \frac{2007+2007}{2007} = \frac{2 \times 2007}{2007} = 2$$

Q42 (2) - Integer Type

$$\text{Given, } \int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx = \mu e^x \left(\frac{1+x}{1-x}\right)^\lambda + C$$

$$\text{Let } I = \int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx$$

$$\Rightarrow I = \int e^x \left[\frac{1}{(1-x)\sqrt{1-x^2}} + \sqrt{\frac{1+x}{1-x}} \right] dx$$

$$\text{As, } \frac{d}{dx} \left[\sqrt{\frac{1+x}{1-x}} \right] = \frac{1}{(1-x)\sqrt{1-x^2}}$$

$$\therefore I = e^x \sqrt{\frac{1+x}{1-x}} + C$$

$$\Rightarrow I = e^x \left(\frac{1+x}{1-x}\right)^{1/2} + C$$

$$\Rightarrow \mu = 1, \lambda = \frac{1}{2}$$

$$\therefore 2\lambda + \mu = \left(2 \times \frac{1}{2} + 1\right) = 2$$

Q43 (2) - Integer Type

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Consider the integral

$$\begin{aligned} I &= \int \frac{dx}{x^n(1+x^n)^{1/n}}, n \in N = \int \frac{dx}{x^{n+1}\left(1+\frac{1}{x^n}\right)^{1/n}} \\ &= \int (1+x^{-n})^{-\frac{1}{n}} x^{-n-1} dx = \int -\frac{1}{n} \int (1+x^{-n})^{-1/n} d(1+x^{-n}) \\ &= -\frac{1}{n} \frac{(1+x^{-n})^{-\frac{1}{n}+1}}{\left(-\frac{1}{n}+1\right)} + K = \frac{1}{1-n} \left(1-\frac{1}{x^n}\right)^{1-\frac{1}{n}} + K \\ &= \frac{1}{1-n} (1+x^{-n})^{\frac{n-1}{n}} + K \end{aligned}$$

$$\text{Now, } \int \frac{dx}{x^n(1+x^n)^{1/n}} = \frac{1}{1-n} (1+x^{-n})^{\frac{n-1}{n}} + K$$

Put $n = 2010$, then

$$\begin{aligned} \int \frac{dx}{x^{2010}(1+x^{2010})^{\frac{1}{2010}}} &= \frac{1}{1-2010} (1+x^{-2010})^{\frac{2010-1}{2010}} + K \\ &= \frac{1}{-2009} (1+x^{-2010})^{\frac{2009}{2010}} + K \\ \Rightarrow a &= 2009, b = 2010, c = 2011 \end{aligned}$$

$$\text{Hence, } \frac{a+b-c}{1004} = \frac{2009+2010-2011}{1004} = \frac{2008}{1004} = 2$$

Q44 (1) - Integer Type

$$\text{Let } I = \int \frac{dx}{(1+\sqrt{x})^{2010}} = \int \frac{\sqrt{x}}{\sqrt{x}(1+\sqrt{x})^{2010}} dx$$

$$\text{Put } 1+\sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$\Rightarrow dx = 2(t-1)dt$$

$$\therefore I = \int \frac{2(t-1)}{t^{2010}} dt$$

$$= 2 \int \left[\frac{1}{t^{2009}} - \frac{1}{t^{2010}} \right] dt$$

$$= 2 \left[\frac{1}{2009 \cdot t^{2009}} - \frac{1}{2008 \cdot t^{2008}} \right] + C$$

$$\therefore \alpha = 2009, \beta = 2008$$

$$\text{Then, } |\alpha - \beta| = 1$$

Q45 (2) - Integer Type

$$\text{Let } I = \int \frac{dx}{x^{1/2} \cdot (1+x^2)^{5/4}}$$

$$= \frac{dx}{x^{1/2} \cdot x^{5/2} \cdot (x^{-2}+1)^{5/4}} = \int \frac{dx}{x^3 \cdot (1+x^{-2})^{5/4}}$$

$$\text{Put } 1+x^{-2} = t \Rightarrow \frac{-2}{x^3} dx = dt$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\begin{aligned} \therefore I &= \frac{-1}{2} \int \frac{dt}{t^{5/4}} \\ &= -\frac{1}{2} \cdot \frac{1}{t^{1/4} \cdot (-1/4)} = \frac{2}{(1+x^2)^{1/4}} \\ &= \frac{2\sqrt{x}}{(1+x^2)^{1/4}} + C \\ \therefore A &= 2 \end{aligned}$$

Q46 (5) - Integer Type

$$\begin{aligned} \text{Let } I &= \int \frac{2x+3}{(x^2+3x)(x^2+3x+2)+1} dx \\ \text{Put } x^2+3x &= t \Rightarrow (2x+3)dx = dt \\ \therefore I &= \int \frac{dt}{t(t+2)+1} = \int \frac{dt}{(t+1)^2} \\ &= C - \frac{1}{t+1} = C - \frac{1}{x^2+3x+1} \\ \Rightarrow a &= 1, b = 3, c = 1 \\ \therefore a + b + c &= 5 \end{aligned}$$

Q47 (4) - Integer Type

$$\begin{aligned} &\int (\sqrt{\tan x} + \sqrt{\cot x}) dx \\ &= \sqrt{\lambda} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{\mu}} \right) + C \\ \text{Now } \int (\sqrt{\tan x} + \sqrt{\cot x}) dx &= \int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}} dx \\ &= \sqrt{2} \int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \sqrt{2} \int \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx \\ &= \sqrt{2} \int \frac{dt}{1-t^2}, \text{ where } t = \sin x - \cos x \\ &= \sqrt{2} \sin^{-1}(t) + C \\ &= \sqrt{2} \tan^{-1} \frac{t}{\sqrt{1-t^2}} + C \\ &= \sqrt{2} \tan^{-1} \left(\frac{\sin x - \cos x}{\sqrt{\sin 2x}} \right) + C \\ &= \sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C \\ \Rightarrow \lambda &= 2, \mu = 2 \\ \therefore \lambda + \mu &= 2 + 2 = 4 \end{aligned}$$